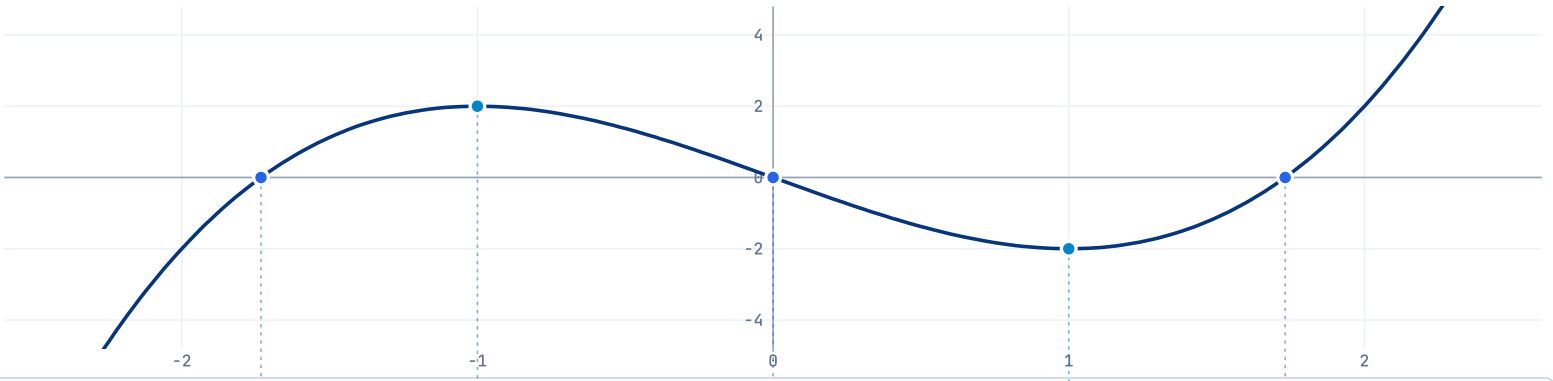
ONE FUNCTION, EVERY FEATURE

Profile of a cubic

$$f(x) = x^3 - 3x = x(x^2 - 3)$$

3 X-INTERCEPTS · 2 LOCAL EXTREMA · ODD SYMMETRY



SIGN OF F	−	+	−	+
DIRECTION	increasing ►	decreasing		increasing ►

KEY POINTS

2 LOCAL MAXIMUM
$(-1, 2)$

1 X-INTERCEPTS
$x = -\sqrt{3},\ 0,\ \sqrt{3}$

3 LOCAL MINIMUM
$(1, -2)$

GLOBAL

DOMAIN
$(-\infty, \infty)$
a polynomial accepts every real input

RANGE
$(-\infty, \infty)$
the two arms run off without bound

Y-INTERCEPT
$(0, 0)$

INCREASING ON
$(-\infty, -1)$ and $(1, \infty)$

DECREASING ON
$(-1, 1)$

SYMMETRY
odd: $f(-x) = -f(x)$
the graph lands on itself after a half turn about the origin

PERIODICITY
none

END BEHAVIOUR
$f(x) \to -\infty$ as $x \to -\infty$; $f(x) \to +\infty$ as $x \to +\infty$

DISCONTINUITIES
none